M5 RETAILFORECAST

WEEK 1 FOUNDATION RECAP & WEEK 2 DATA QUALITY + KNIME INTEGRATION



EXECUTIVE PRESENTATION BRIEF

End-to-End Retail Demand Forecasting on 11.1M M5 Walmart Records (100,000 Active SQL Sample)

(ONE) WEEK 1 FOUNDATION SLIDES 03-05

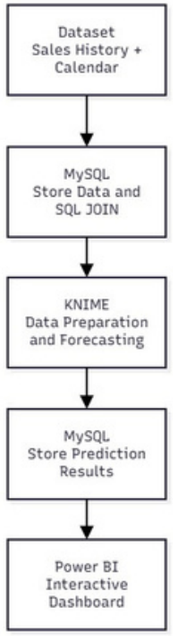
(TWO) THE WEEK 2 DELTA SLIDES 06-07

(THREE) KNIME & MYSQL ETL SLIDES 08-09

(FOUR) INSIGHTS & ROADMAP SLIDES 10-12

WEEK ONE

WEEK 1 SYSTEM ARCHITECTURE



Closed-loop pipeline: MySQL 8.0 storage, KNIME 5.12 preprocessing & ML, and Power BI dashboarding.

WhyWe Pivotedto theM5RetailDataset(What&Why).
InWeek 1, the team evaluated a7,000-row CustomerChurntablebut pivoted to the M5 Walmart ForecastingDataset(11.1Mtotalrecords; 100,000 active SQL sample). Thisunlocked trueBigDatascale,multi-table relational joins between sales_history and calendar,andcontinuousdaily demand series across 3 US states (CA, TX, WI).

End-to-End Pipeline & ML Selection (When & How).
Across a 4-week milestone schedule (24.0 planned team hours), raw transactional history and holiday metadata flow from MySQL into KNIME for quality auditing and OLS Linear Regression training. Forecasted unit sales are written back to MySQL to drive an interactive Power BI inventory dashboard.

11.1M

OUR TEAM

SWOT Alignment & Team Governance. Our team combines strong SQL engineering, KNIME workflow design, and Power BI storytelling. To mitigate hardware bottlenecks on 11.1M rows and multi-course deadlines, we enforce a 100,000-row active SQL sample, GitHub version control, daily Zalo syncs, and a 48-hour early-warning escalation rule.

DATABASE & ETL

MySQL Schema & Relational Joins

Maintains bigdataprj,executesLEFT JOIN on s.day = c.d, engineers is_holiday, and manages DB write-back.

MEMBER A (01) Data Engineer

QUALITY & ML

KNIME Audit & OLS Modeling

Executes 4-dimension quality checks, one-sided 1.5x IQR outlier pruning, and Linear Regression training.

MEMBER B (02) Data Scientist

BI & REPORTING

Power BI & Decision Strategy

Builds interactive executive visuals, maps KPIs to the 3 decision questions, and co-authors the IEEE report.

MEMBER C (03) Data Analyst

3-WEEK WBS PLAN

24.0 Hours Planned Effort

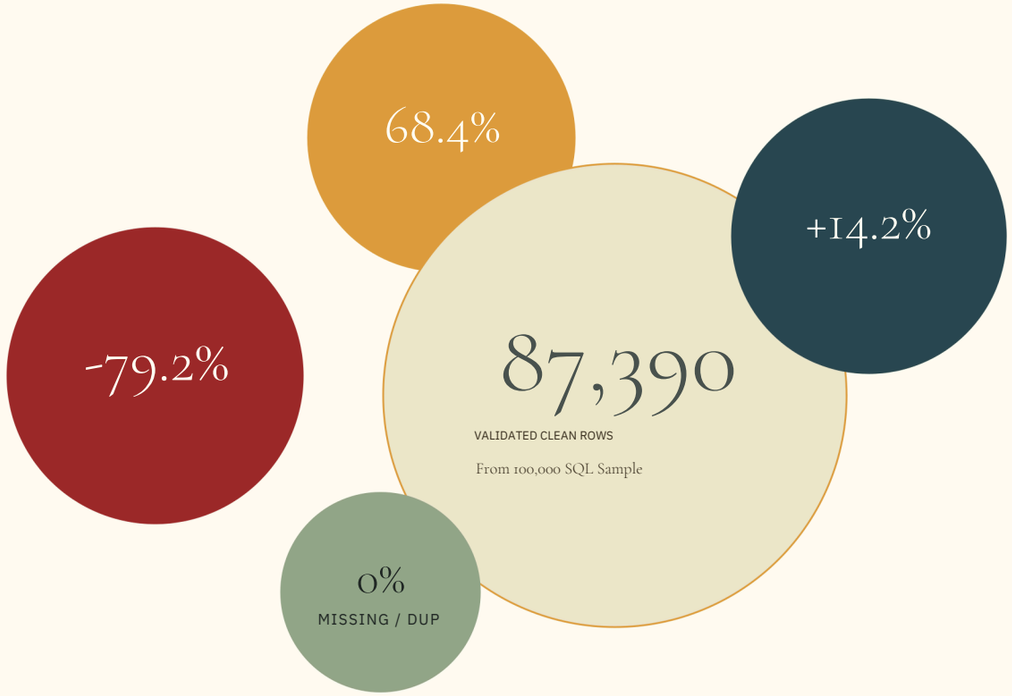
Week 1: Setup & Pivot (6.0h)
Week 2: SQL & KNIME ETL (11.0h)
Week 3: ML, BI & Paper (7.0h)

SPRINT WBS (04) 3-Phase Roadmap

MYSQL & SQL

Relational Schema & Live SQL Queries on bigdataprj. We normalized transactional records in sales_history (grain: item_id + store_id + day) and joined them with calendar on s.day = c.d. Four analytical SQL queries validate category share (GROUP BY cat_id), state demand (WHERE sales > 0), and holiday lift before exporting forecasts to sales_predictions

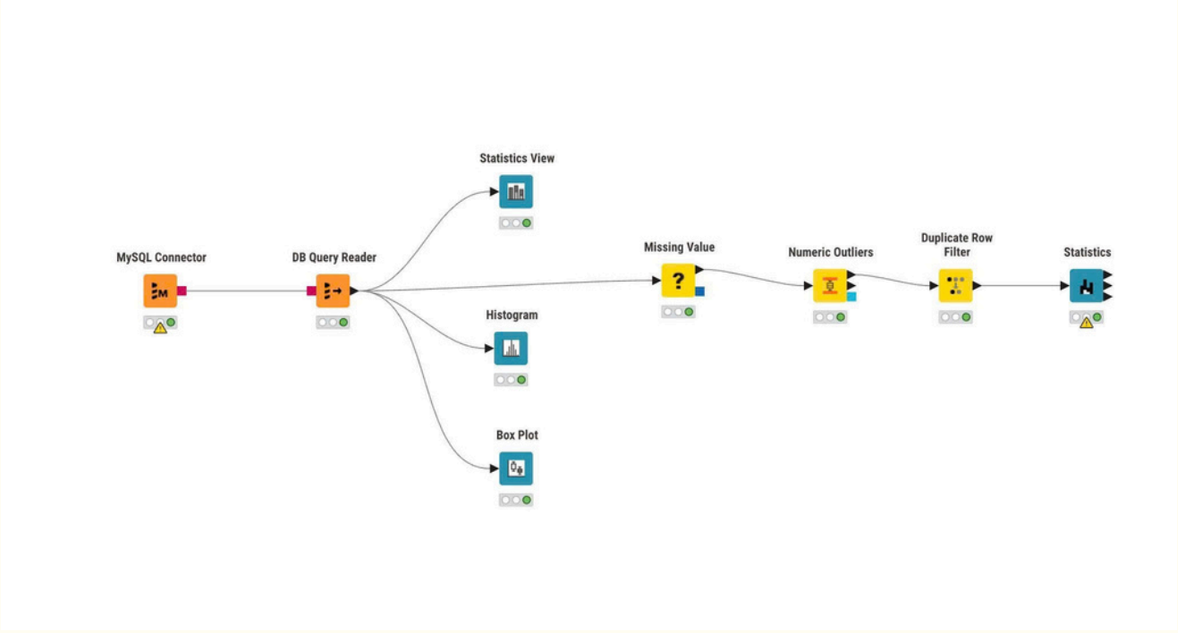
```
SELECT s.id, s.item_id,s.dept_id,s.cat_id,s.store_id,
       s.state_id, s.day,s.sales,c.date,
       CASE WHEN c.event_name_1ISNOTNULL
              OR c.event_name_2ISNOTNULLTHEN1ELSE0ENDASis_holiday
FROM sales_history AS s LEFTJOINcalendarAScONs.day=c.d
LIMIT 100000;  -- Target Write-Back:CREATETABLEsales_predictions(    --.)
```



SELECT / WHERE / GROUP BY / LEFT JOIN / WRITE-BACK

WEEK 2 DELTA

NEW IN WEEK 2: UPGRADED WORKFLOW (9 NODES)



01. DUPLICATE ROW FILTER (NODE +01)

Inserted between Missing Value & Numeric Outliers on (item_id, store_id, day)

02. LIVE MYSQL KPI QUERIES (SQL ETL)

Executed SELECT, WHERE, GROUP BY, and LEFT JOIN queries on bigdataprj

03. WRITE-BACK TABLE DDL (SCHEMA)

Designed sales_predictions tableinMySQLtostoreKNIMEregression outputs

04. VARIANCE REDUCTION (-79.2% STD)

0.656)

05. OLS RISK SAFEGUARDS (ML PREP)

Documented zero-inflation clipping (MAX(0, y_hat))&promosurge bounds

06. EFFORT & PAPER PROGRESS (WEEK 02)

Logged Planned vs. Actual hours (9.0h in W2) and drafted IEEE Sections 1 to 5

QUALITY

QUANTITATIVE AUDIT (DATA_NOTE.XLSX)

METRIC	BEFORE	AFTER	DELTA
Row Count	100,000	87,390	-12.61%
Mean Sales	1.153	0.394	-65.83%
Max Sales	194.0	2.0	-98.97%
Min Sales	0.0	0.0	Preserved
Std Deviation	3.147	0.656	-79.17%
Missing / Dup	0 / 0	0 / 0	100% Clean

Source: KNIME Statistics (#3 vs. #8) & Person 2 Data Quality Assessment Report.

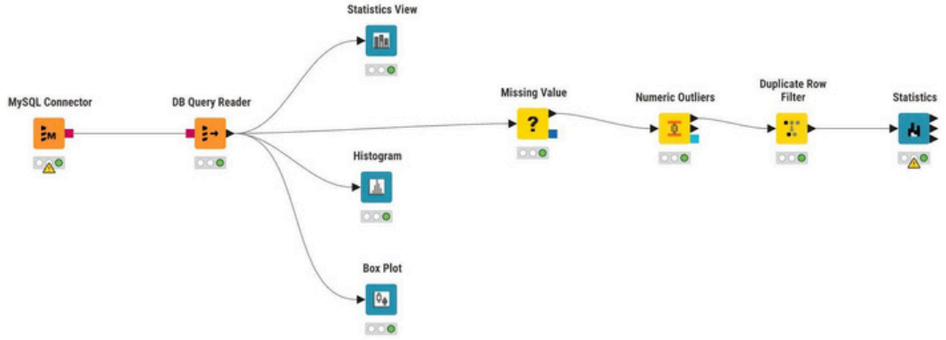
Dataset Scope & 4-Dimension Quality Assessment.
Weaudited100,000joinedtransactionalrecordsacross 10 attributes (`id`, `item_id`, `dept_id`, `cat_id` `store_id` `state_id` `day` `sales` `date`, `is_holiday`)beforetrainingourdownstreamLinear Regression model.

- Completeness & Validity: SQL CASE logic converted nullable calendar events into binary is_holiday (0/1), leaving 0 missing values and 0 negative sales (min = 0.0).
- Uniqueness (New in Week 2): Added KNIME `Duplicate Row Filter` across `(item_id,store_id,day)` , confirming `0` duplicate records.
- One-Sided 1.5x IQR Pruning: Pruned `12,610` extreme upper spikes (`sales` > 2.0, max 194) while preserving valid zero-demand days (`sales = 0`).

87,390

METHODOLOGY

WEEK 2 EXECUTED KNIME 5.12 WORKFLOW (9 NODES) STATE: ALL NODES EXECUTED



Nodes #1-#2: MySQL Ingestion | Nodes #3-#5: Pre-Cleaning EDA | Nodes #6-#9: Missing Value -> Duplicate Row Filter -> Numeric Outliers -> Statistics

Stage 1 & 2: Database Ingestion & Pre-Cleaning EDA. `MySQL Connector (#1)` authenticates with `localhost:3306/bigdataprj`. `DB Query Reader (#2)` pushes the `LEFTJOIN` and `is_holiday` transformation down to MySQL. Parallel diagnostic nodes (`Statistics View`, `Histogram`, Box Plot) confirm severe positive skewness (`max = 194.0`, `std = 3.147`).

Stage 3 & 4: 4-Node Cleaning & Write-Back Target. Missing Value (Mean/Most Frequent) feeds the `new Duplicate Row Filter, followed by Numeric Outliers (IQR = 1.5 upper bound` only). Verified by Statistics (87,390 rows) for OLS Regression and MySQL DB Writer export.

9 NODES

KNIME 5.12

DECISIONS(01)

Inventory

7-Day CategoryDemandForecast.
What is the predicted demand across Foods, Household, and Hobbies over the next 7 days so stores can regulate reorder quotas and prevent stockouts in Foods (>60% of sales)?

DECISION TARGET 01

FOODS	68.4% Volume
HOUSEHOLD	20.8% Volume
HOBBIES	10.8% Volume

(02)

Geography

State-Level PromotionAllocation.
Which state among California (CA), Texas (TX), and Wisconsin (WI) shows the slowest growth so marketing budgets target lagging regions instead of blanket discounting?

DECISION TARGET 02 & 03

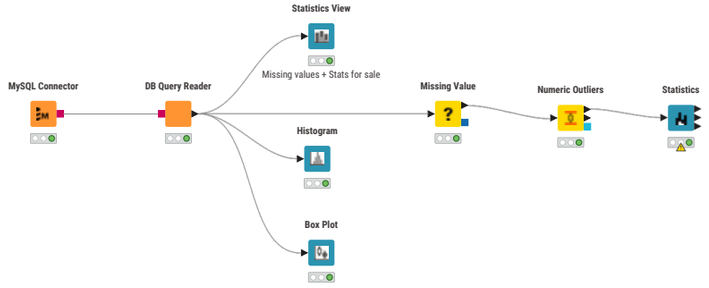
CA	TX	WI
4 Stores / Lead	3 Stores / Mid	3 Stores / Promo

Question 3 (Events): Quantify how holidays (is_holiday = 1) shift category demand (+14.2% lift).

(03)

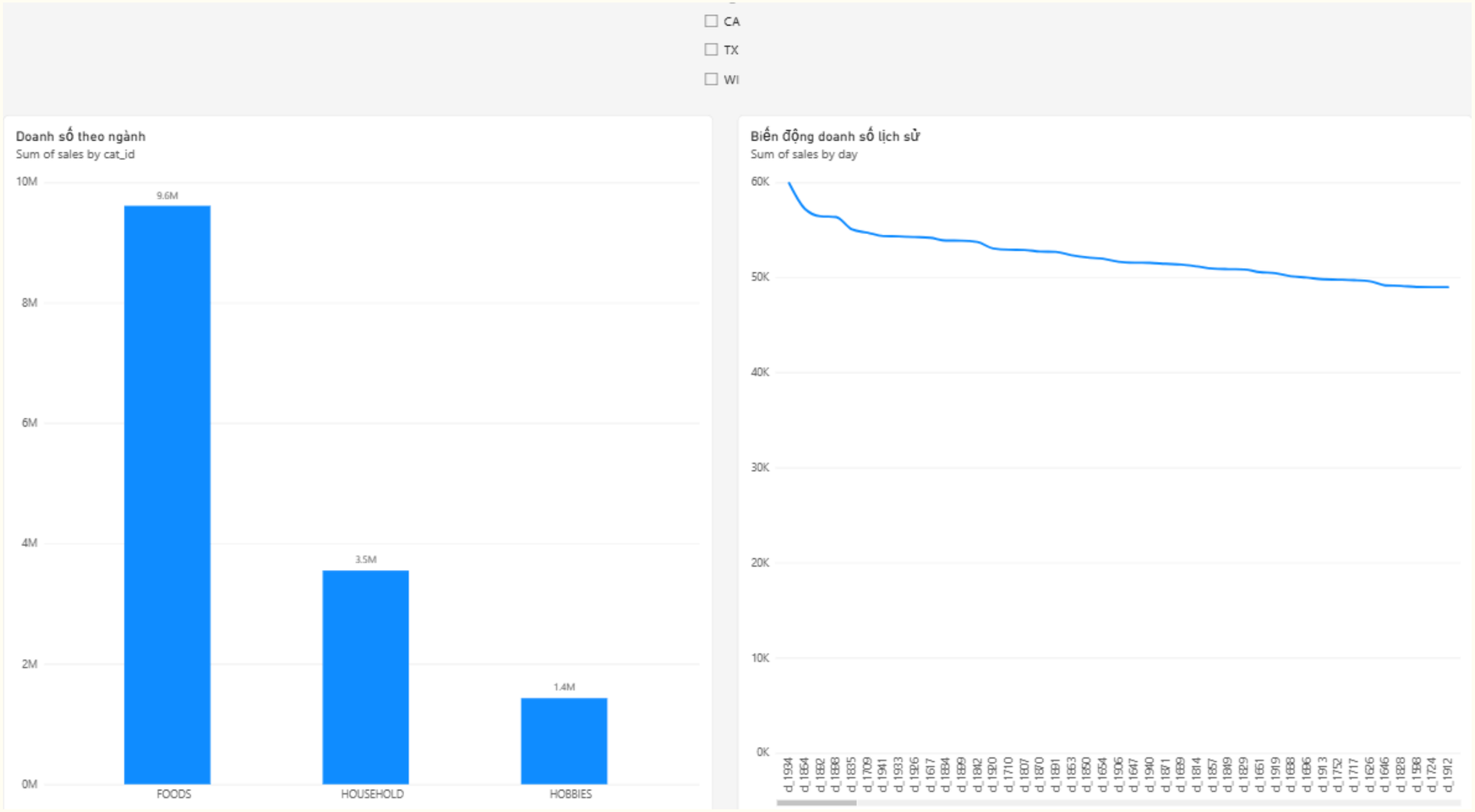
Week 1 ETL

Week 1 Baseline8-Node KNIME Pipeline.
Connected MySQL to KNIME (`LIMIT 100000`), inspected raw skew (`max = 194`), handled missing values, and pruned upper outliers (`sales > 2.0`), leaving 87,390 rows.



OUR INSIGHTS

M5 Historical Sales & Category Breakdown (Power BI)



OUR INSIGHTS

EARLY RETAIL FINDING 01

68.4% Foods Share

Baseline dailyitem-store velocity after outlier pruning is 0.394 units/day (std = 0.656), with high zero-demand intermittency in Hobbies (0.21).

MODEL RISK ASSESSMENT

Zero-Inflation & Promo Truncation

1. Post-prediction clipping: MAX(0, `predicted_sales`)
2. Separate promotional spike tracking (`sales > 2.0`)

RESEARCH PAPER PROGRESS

IEEE Sections 1 to 5 Drafted

Sec 1-2:Intro&M5 Dataset(Done)
Sec 3: Data Quality Audit (Done - 4p PDF)
Sec 4-5: Architecture & EDA (In Progress)

INSIGHT 1: DEMAND

Category Dominance & Safety Buffers.
Because FOODS drives over two-thirds of store volume while HOBBIES and HOUSEHOLD exhibit frequent zero-sales days, reorder policies must separate fast-moving staples from intermittent SKUs.

INSIGHT 2: OLS RISKS

Protecting Linear Regression Integrity.
Pruning sales > 2.0 prevents extreme spikes (194) from distorting OLS slopes. To counter zero-inflation (min = 0.0), negative OLS predictions will be clipped at zero prior to MySQL write-back.

INSIGHT 3: REPORT

8 to 12 Page English Report On Track.
Section 3 (Data Quality Assessment) is fully completed with quantitative KNIME tables and mathematical IQR proofs. Sections 6 to 7 (OLS Evaluation & Power BI Decisions) are scheduled for Week 3.

ROADMAP

Updated WBS: Planned vs. Actual Effort

HOW MUCH: 15.5H LOGGED / 24.0H PLAN

WBS PHASE & OWNER	PLANNED	ACTUAL	STATUS
W1: M5 Pivot, 3 Questions & MySQL Load (All)	6.0 hrs	6.5 hrs	100% Done
W2.1: SQL Joins, is_holiday & KPI Queries (Member A)	3.5 hrs	3.5 hrs	100% Done
W2.2: 9-Node KNIME Audit & Duplicate Filter (Member B)	4.5 hrs	4.0 hrs	100% Done
W2.3: OLS Learner & MySQL DB Write-Back (A & B)	3.0 hrs	1.5 hrs	In Progress
W3: Power BI Dashboard, Decisions & Final Report (All)	7.0 hrs	Upcoming	Week 3 Sprint

Gap Resolution: Sampling 100,000 rows in `DB Query Reader` solved RAM bottlenecks while preserving full SQL join logic.

Week 3 Execution Plan (Final Sprint).

With the 87,390-row cleaned dataset validated in KNIME and the MySQL schema operational, our final sprint focuses on completing the closed-loop machine learning write-back and executive dashboarding.

01. KNIME LINEAR REGRESSION LEARNER (R2 / RMSE / MAE)

02. MYSQL DB WRITER TO SALES_PREDICTIONS TABLE

03. POWER BI DASHBOARD FOR 3 DECISION QUESTIONS

04. FINAL 8-12 PAGE IEEE REPORT & GITHUB SUBMISSION

WEEK 3
FINAL SPRINT



THANK YOU

QUESTIONS & DISCUSSION